How-To Library

Informatica

Sizing and Tuning Guidelines for Enterprise Data Lake 10.2.1

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Abstract

You can size and tune your Enterprise Data Preparation deployment for optimum performance. This article provides sizing guidelines for Informatica domain nodes and Hadoop clusters based on your Enterprise Data Preparation deployment type. The article also offers recommendations for tuning the application services associated with Enterprise Data Preparation.

Supported Versions

- Enterprise Data Lake 10.2.1

Table of Contents

Overview. 2
Enterprise Data Preparation Deployment Types. 3
Recommendations for Domain Nodes. 4
Rules and Guidelines for Sizing Informatica Nodes. 5
Recommendations for Hadoop Clusters. 5
Rules and Guidelines for Sizing Hadoop Clusters. 6
Informatica Application Services Configuration Recommendations. 7
Enterprise Data Lake Service Configuration Recommendations. 7
Data Preparation Service Configuration Recommendations. 8
Model Repository Service Configuration Recommendations. 11
Data Integration Service Configuration Recommendations. 11
Catalog Service Configuration Recommendations. 12
Profiling Performance Recommendations. 13

Overview

Informatica recommends that you tune your Enterprise Data Lake deployments for optimal performance. Tuning improves the performance of all Enterprise Data Lake operations, from previewing data assets and preparing worksheets to publishing assets to the data lake.

To tune Enterprise Data Lake, you configure or size the following components:

- The Informatica domain nodes on which the Informatica application services run.
- The Hadoop clusters on which Enterprise Data Lake jobs run.
- The Informatica application services associated with Enterprise Data Lake.

The following table describes the Enterprise Data Lake operations and the application services and components involved in each operation:

Operation/ Component	Model Repository Service	Data Integration Service	Catalog Service	Enterprise Data Lake Service	Data Preparation Service	Hadoop Cluster
Import Data Asset	Yes	Yes	Yes	Yes	No	Yes
File Upload	No	Yes	Yes	Yes	No	Yes
Data Asset Preview	No	No	Yes	Yes	No	No
Project Management	Yes	No	No	Yes	No	No
Addition of Data Asset to a Project	Yes	No	No	Yes	No	No
Worksheet Preparation	Yes	No	No	Yes	Yes	No
Worksheet Publication	Yes	Yes	Yes	Yes	Yes	Yes
Export Data Asset	Yes	Yes	Yes	Yes	No	Yes
Search Catalog	No	No	Yes	Yes	No	No

Enterprise Data Lake runs jobs on various execution engines in a Hadoop cluster. The Hive engine runs MapReduce jobs. The Blaze engine runs Informatica tasks and tasklets. The Spark engine runs Spark jobs.

The following table lists the Enterprise Data Lake operations and the corresponding execution engine for each operation:

Operation/Component	Hive Engine	Blaze Engine	Spark Engine
Worksheet Publication	Yes	Yes	No
Random Sampling	Yes	No	Yes
File Upload	Yes	No	No
Import Data Asset	Yes	Yes	No
Export Data Asset	Yes	Yes	No

Enterprise Data Preparation Deployment Types

To help you size and tune Enterprise Data Preparation, Informatica defines four Enterprise Data Preparation deployment types. Each deployment type is based on factors such as the number of catalog objects, the operational data volume, the total data volume, and the number of active users. The operational data volume is the amount of data

you typically process at a particular point in time, such as the amount of data you typically process at 9:00 PM each Monday.

The following table describes the number of catalog objects, the operational data volume, the total data volume, and the number of active users for each deployment type:

Deployment Type	Number of Catalog Objects	Total Data Volume	Operational Data Volume	Number of Active Users
Sandbox	100,000	100 GB	1 GB	1
Basic	1 million	10 TB	100 GB	5
Standard	20 million	100 TB	1 TB	10
Advanced	50 million	1 PB	10 TB	50 or more

Recommendations for Domain Nodes

The number of nodes required for optimum Enterprise Data Lake performance depends on your deployment type.

For sandbox and basic deployment types, you might run all of the application services required by Enterprise Data Lake on a single node. For standard and advanced deployment types, you might distribute the application services across multiple nodes for better performance.

Node Sizing Recommendations

The following table describes the node sizing recommendations for each Enterprise Data Lake deployment type:

Deployment Type	Total Virtual Cores	Total RAM	Maximum User Processes	File Descriptor Size	Number of Nodes
Sandbox	16	32 GB	1024	16K	1
Basic	24	32 GB	3000	32K	1
Standard	36	64 GB	8000	32K	2
Advanced	96	128 GB	12000	64K	4

Application Service Distribution Recommendations

The following table describes recommendations for distributing application services across multiple nodes in standard or advanced deployment types:

Deployment Type	Node 1	Node 2	Node 3	Node 4
Standard	- Model Repository Service - Data Integration Service - Enterprise Data Lake Service - Data Preparation Service - Catalog Service - Analyst Service - Scheduler Service	- Monitoring Model Repository Service - Data Integration Service - Data Preparation Service - Content Management Service - Search Service	Not applicable.	Not applicable.
Advanced	- Data Integration Service - Enterprise Data Lake Service - Data Preparation Service - Catalog Service - Scheduler Service	- Data Integration Service - Monitoring Model Repository Service - Data Preparation Service - Content Management Service	- Model Repository Service - Data Integration Service - Data Preparation Service - Search Service	- Data Integration Service - Data Preparation Service - Analyst Service

Configure the Enterprise Data Lake Service and the Catalog Service on gateway nodes. Configure the Catalog Service on a single node, because failover is not supported for the Catalog Service.

Rules and Guidelines for Sizing Informatica Nodes

Consider the following guidelines when you size your Informatica nodes:

- The minimum disk space required for each node is 100 GB.
- The recommendations assume that profiling jobs are not run concurrently.

Recommendations for Hadoop Clusters

Use the recommendations in this section to properly size Hadoop clusters for your Enterprise Data Lake deployment type.

The Data Integration Service runs jobs in a Hadoop cluster when Enterprise Data Lake users upload, sample, or publish data assets to the Data Lake. Enterprise Data Catalog runs scanner jobs to extract metadata from data assets, and then stores the metadata in the catalog.

You can run all Data Integration Service jobs and Enterprise Data Catalog jobs on a single Hadoop cluster. For improved performance, you can dedicate a Hadoop cluster to run Data Integration Service jobs, and dedicate another Hadoop cluster to run Enterprise Data Catalog jobs.

The recommendations are based on the assumption that jobs run with eight map waves at the operational volume and with the specified number of active users noted for each deployment type.

Sizing Recommendations for Hadoop Clusters that Run All Jobs

The following table describes the sizing recommendations for a single Hadoop cluster that runs all jobs for each Enterprise Data Lake deployment type:

Deployment Type	Total Number of Virtual Cores for YARN	Total Memory for YARN
Sandbox	24	32 GB
Basic	64	96 GB
Standard	528	576 GB
Advanced	5112	5.2 TB

Sizing Recommendations for Hadoop Clusters Dedicated to Data Integration Service Jobs

The following table describes the sizing recommendations for Hadoop clusters dedicated to running Data Integration Service jobs for each Enterprise Data Lake deployment type:

Deployment Type	Total Number Virtual Cores for YARN	Total Memory for YARN
Sandbox	16	32 GB
Basic	64	64 GB
Standard	544	544 GB
Advanced	5056	5.1 TB

Sizing Recommendations for Hadoop Clusters Dedicated to Enterprise Data Catalog Jobs

The following table describes the sizing recommendations for Hadoop clusters dedicated to running Enterprise Data Catalog jobs for each Enterprise Data Lake deployment type:

Deployment Type	Total CPU Cores for YARN	Total Memory for YARN
Sandbox	8	24 GB
Basic	8	24 GB
Standard	24	72 GB
Advanced	48	144 GB

Rules and Guidelines for Sizing Hadoop Clusters

Consider the following rules and guidelines when you size a Hadoop cluster:

- Set the file descriptor limit to 64K on each data node in the cluster.
- The Hadoop cluster CPU core and memory requirements include the Catalog Service requirements.

- The Blaze Grid Manager uses one core and 4 GB of memory for each node. For example, if you have a cluster with 14 nodes, the Blaze Grid Manager uses 14 cores and 14 × 4 = 56 GB memory. The grid monitor uses only one core and 1 GB of memory across the entire Hadoop cluster.
- The recommendations assume that profiling jobs are not run concurrently.

Informatica Application Services Configuration Recommendations

Configure the Informatica application services that are associated with Enterprise Data Lake for optimum performance.

You can configure the following application services:

- Enterprise Data Lake Service
- Data Preparation Service
- Model Repository Service
- Data Integration Service
- Catalog Service

You use Informatica Administrator (the Administrator tool) to configure the properties for each application service.

Enterprise Data Lake Service Configuration Recommendations

The Enterprise Data Lake Service is the application service that runs the Enterprise Data Lake application in the Informatica domain.

Heap Size Recommendations

Specify the maximum heap size to allocate to the Enterprise Data Lake Service according to the Enterprise Data Lake deployment type. The default maximum heap size is 512 MB.

The following table describes the recommended heap size for each Enterprise Data Lake deployment type:

Deployment Type	Maximum Heap Size
Sandbox	512 MB
Basic	1 GB
Standard	2 GB
Advanced	4 GB

Hive Table Storage Format

Set the Hive Table Storage Format property to columnar storage formats like Optimized Row Columnar (ORC) or Parquet for better storage efficiency and improved performance.

Data Asset Recommendation Options

Use alternate and additional data assets for a project based on existing data assets to improve productivity. By default, the number of recommendations to display is 10. The recommendations include requests sent to the Catalog Service and you need to carefully set the number of recommendations.

Update Hive Table Statistics

Sampling tables with Hive statistics spawns a query without using the MapReduce reduce task. As a result, the YARN container requires fewer resources than when sampling a table without statistics.

Update Hive table statistics at regular weekly or biweekly intervals for better sampling performance.

Importing Data into the Data Lake

When you import a data asset into the data lake, Enterprise Data Lake triggers a Blaze job to perform the import activity.

The Blaze engine uses Sqoop to connect to the source system that contains the data asset to import. The Blaze engine spawns four map tasks by default to read the data. You can use the *-num-mappers* or *-m sqoop* Sqoop parameters to increase the number of map tasks to improve import job performance.

If you import a table that does not contain a primary key, profile the table before you import the table into the data lake.

Scheduling Export, Import, and Publish Activities

Exporting, importing, and publishing data assets consumes Informatica node and Hadoop cluster resources. If you schedule 10 or more export, import, or publish activities, schedule the activities to run during non-business hours or low demand times.

Apache Zeppelin Options

Enterprise Data Lake integrates with the Apache Zeppelin notebook application. You can use Zeppelin to visualize published data assets.

While running Zeppelin queries using the Spark interpreter, Zeppelin utilizes all available YARN resources, which might result in contention with Enterprise Data Lake jobs submitted to the same cluster.

You can limit resource consumption by modifying the ZEPPELIN_JAVA_OPTS variable in the `$ZEPPELIN_HOME/conf/zeppelin-env.sh` file. Set the following properties in the variable:

- `spark.executor.memory`
- `spark.executor.cores`
- `spark.executor.instances`
- `spark.yarn.queue`

Using a separate YARN queue with restricted capacity and enabling Spark dynamic allocation is also highly recommended.

Data Preparation Service Configuration Recommendations

The Data Preparation Service manages data preparation within the Enterprise Data Lake application.

Heap Size Recommendations

Set the maximum heap size to a higher value for larger concurrency. The default maximum heap size is 8 GB.

Configure the Maximum Heap Size property based on your Enterprise Data Lake deployment type. Allocate a minimum of 512 MB per user.

The following table describes the maximum heap size recommendations for each Enterprise Data Lake deployment type:

Deployment Type	Maximum Heap Size Value
Sandbox	2 GB
Basic	4 GB
Standard	8 GB
Advanced	20 GB

The recommendations assume that each project contains five to 10 worksheets, with 10 to 50 columns in each worksheet. If Enterprise Data Lake users typically create projects with more worksheets, and if users create worksheets with more columns, you might want to increase the heap size to improve performance.

Local and Durable Storage Disk Space Requirements

When a user opens or refreshes a worksheet in Enterprise Data Lake, the Data Preparation Service stores three copies of the worksheet in the local storage and in the durable storage. The default size of the sample data in a worksheet is 50,000 rows.

Use the following formula to estimate the disk space requirements for the local storage and the durable storage:

```
rows in an opened worksheet (default 50,000) × the number of concurrent users × 3 (copies of
the worksheet in the local and durable storage) + 2 GB (additional disk space)
```

Pivot and Unpivot Operations

Performing a pivot operation or an unpivot operation involving 100 or more columns might increase JVM consumption by the Data Preparation Service. Memory consumption might also increase when five or more concurrent users perform pivot and unpivot operations.

Tune the Data Preparation Service heap size per the heap size recommendations to avoid performance impacts.

Node Allocation for Data Preparation Service on a Grid

You can run the Data Preparation Service on a grid. Running the Data Preparation Service on a grid enables you to add additional nodes in the domain to accommodate higher concurrency when needed. Informatica recommends that you run the Data Preparation Service on a grid for standard and advanced Enterprise Data Lake deployment types.

The following table describes the recommended number of active users and number of nodes when running the Data Preparation Service on a grid:

Deployment Types	Nodes Running the Data Preparation Service
Sandbox	1
Basic	1
Standard	2
Advanced	4 or more

Sampling Recommendations

Sample Size

The Default Data Preparation Sample Size property determines the number of sample rows loaded into worksheets during data preparation.

The value of the Default Data Preparation Sample Size property has an impact on the performance and user experience when a user loads or refreshes the worksheets in the data preparation page. Setting the property to a higher value might result in higher memory usage by the Data Preparation Service.

The default property value is 50,000 rows. The minimum value is 1000, and the maximum value is one million.

Random Sampling

Enterprise Data Lake supports random sampling of data. Use this option if you have high volumes of data, or if the input involves multiple splits.

When a user selects the Random Sampling option when sampling less than 1 GB of data, Enterprise Data Lake spawns a MapReduce job on the Hadoop cluster to load the sample data.

If a user samples more than 1 GB of data, Enterprise Data Lake spawns a job using the default processing engine on the Hadoop cluster. Configure the Hadoop cluster to use the Spark engine as the default processing engine, because the Spark engine performs better than MapReduce when performing random sampling on large amounts of data.

Data Preparation Repository Recommendations

The Data Preparation Service stores recipe and mapping metadata in the Data Preparation repository. You can use a MySQL database, a MariaDB database, or an Oracle database as the Data Preparation repository database.

The following table describes the CPU and memory requirements for a MySQL database server or a MariaDB database server:

Deployment Type	CPU Cores	Memory	Disk Space
Sandbox	1	2 GB	10 GB
Basic	1	2 GB	50 GB
Standard	2	4 GB	100 GB
Advanced	4	8 GB	200 GB

The following table describes the CPU and memory requirements for an Oracle database server:

Deployment Type	CPU Cores	Memory	Disk Space
Sandbox	2	4 GB	10 GB
Basic	2	4 GB	50 GB
Standard	4	6 GB	200 GB
Advanced	8	10 GB	400 GB

Model Repository Service Configuration Recommendations

The Model Repository Service associated with Enterprise Data Lake manages the Model repository that stores project metadata and the mappings created from published data.

Heap Size Recommendations

The following table describes the Model Repository Service maximum heap size recommendations for each Enterprise Data Lake deployment type:

Deployment Type	Maximum Heap Size
Sandbox	1 GB
Basic	1 GB
Standard	2 GB
Advanced	4 GB

Model Repository Recommendations

Enterprise Data Lake writes user activity and project history data to the PO_IDLEVENT table in the Model repository database. Each record in the PO_IDLEVENT table requires an average of 1.5 KB of disk space.

The following table contains estimates on the disk space required based on the number of user activity event records added to the PO_IDLEVENT table:

Records in the PO_IDLEVENT Table	Estimated Storage Required
100,000	150 MB
200,000	300 MB
1 million	1.5 GB
3 million	4.5 - 5 GB

Data Integration Service Configuration Recommendations

The Data Integration Service associated with Enterprise Data Lake performs data integration tasks for Enterprise Data Lake and writes uploaded and prepared data to a Hive table in a Hadoop cluster.

Recommended Data Integration Service Configuration

The following table describes the recommended Data Preparation Service configuration for each Enterprise Data Lake deployment type:

Deployment Type	Maximum Heap Size	Maximum On-Demand Execution Pool Size	Hadoop Batch Execution Size	Native Batch Execution Size
Sandbox	640 MB	10	10	10
Basic	2 GB	10	500	10

Deployment Type	Maximum Heap Size	Maximum On-Demand Execution Pool Size	Hadoop Batch Execution Size	Native Batch Execution Size
Standard	2-4 GB	15	1000	15
Advanced	4-6 GB	75	2000+	30

To improve performance for larger sample sizes and a large number of concurrent Enterprise Data Lake users, you might want to run the Data Integration Service on a grid.

Rules

Applying rules during data preparation impacts resource usage on Data Integration Service nodes. For larger sample sizes, increase the value of the On-Demand Max Execution Pool property. The Maximum On-Demand Execution Pool Size property controls the maximum number of requests that the Data Integration Service can run concurrently.

Use the following formula to determine the ideal value for the On-Demand Max Execution Pool property:

```
number of concurrent mappings x number of concurrent users
```

Follow these guidelines to determine the number of concurrent mappings submitted to the Data Integration Service when applying rules:

- If the number of rows in the data asset and the sampling size is less than or equal to 8192, the number of concurrent mappings submitted can be equal to 1.
- If the number of rows in the data asset and the sampling size is greater than 8192, apply the following formula to determine the number of concurrent mappings submitted:

  ```
  1 + [(number of sampling rows - 8192) / 65536]
  ```

JavaScript Object Notation (JSON) Files

When a user publishes a JSON file, the Data Integration Service uses ORC compression when creating the corresponding table in Hive. You might encounter performance issues when the compression ratio is high.

To improve performance, set the `ExecutionContextOptions.Blaze.HiveStagingCompressionFormat` custom property in the Data Integration Service. Set the custom property value to `NONE`.

Catalog Service Configuration Recommendations

The Catalog Service runs Enterprise Data Catalog, and manages the catalog of information assets in the Hadoop cluster.

Enterprise Data Lake uses Enterprise Data Catalog to catalog and ingest the data lake table metadata and search results. You use the Administrator tool to tune the Catalog Service associated with Enterprise Data Lake.

The Catalog Service uses the Hive scanner to ingest Hive tables to be catalogued and searched in the catalog. You use Catalog Administrator to configure the Hive scanner used by the Catalog Service.

Load Type Property Recommendations

The resource demand and performance from the Hadoop cluster by the Catalog Service depends on the value set for the Load Type property.

The following table describes the recommended values for the Load Type property based on deployment size:

Value	Description
Demo	Set for a one node deployment with up to 100,000 catalog objects.
Low	Set for a one node deployment with up to one million catalog objects.
Medium	Set for a three node deployment with up to 20 million catalog objects.
High	Set for a six node deployment with up to 50 million catalog objects.

Objects include the sum of number of tables and number of columns in all the tables.

Hive Scanner Configuration

You can configure the scanner to extract all schemas or a particular schema. Select a specific schema for better performance. Specify the schema in the Source Metadata section on the Metadata Load Settings tab for the Hive resource in Catalog Administrator.

You can also configure the memory that the Hive scanner process consumes. Configure the Memory property on the **Metadata Load Settings** tab for the Hive resource in Catalog Administrator. Informatica recommends that you set the memory based on the total number of columns that need to be ingested.

The following table describes the recommended values for the Memory property:

Value	Description
Demo	Set for up to 100,000 columns.
Low	Set for up to one million columns.
Medium	Set for up to four million columns.
High	Set for up to 12 million columns.

Spark Dynamic Resource Allocation Configuration

Spark provides a mechanism to dynamically allocate and remove resources based on workload demand. Enabling dynamic allocation is recommended if you have concerns about YARN resource demand, resource constraint, or resource contention, especially where the cluster is shared by multiple applications.

To enable dynamic allocation for Enterprise Data Catalog's Spark application, use the Administrator tool to set the `LdmCustomOptions.ingest.spark.dynamic.allocation.enable` custom property to `true` on the Catalog Service.

Profiling Performance Recommendations

Tuning the profiling parameters is important, because profiling jobs run on the same Hadoop cluster as Enterprise Data Lake.

Blaze Engine for Profiling Jobs

Informatica recommends that you perform profiling using the Blaze engine for best performance.

Incremental Profiling

Incremental profiling is not supported for Hive data sources where incremental changes in data are undetected. Only new tables are detected.

Do not enable incremental profiling for Hive data sources in Enterprise Data Lake if a change in data must be detected for incremental scans.

Profiling Performance Parameters

The profiling warehouse database stores profiling and scorecard results. You use the Administrator tool to configure the profiling warehouse connection in the Data Integration Service properties.

The following table describes the Data Integration Service properties that you can tune to improve profiling performance:

Property	Description
Profiling Warehouse Database	Connection name for the profiling warehouse database. In addition to the profile results, the profiling warehouse holds the persisted profile job queue. Verify that no profile job runs when you change the connection name. Otherwise, the profile jobs might stop running because the profile jobs run on the Data Integration Service where the Profiling Service Module submitted the profile jobs. You set the default value when you create the instance.
Maximum Profile Execution Pool Size	The number of profile mappings that the Profiling Service Module can run concurrently when the Data Integration Service runs on a single node or on a grid. The pool size is dependent on the aggregate processing capacity of the Data Integration Service, which you specify for each node on the **Processes** tab of the Administrator tool. The pool size cannot be greater than the sum of the processing capacity of all nodes. Default is 10.
Maximum Concurrent Columns	The number of columns that a mapping runs in parallel. The default value is optimal for most profiling use cases. You can increase the default value for columns with cardinality lower than the average value. Decrease the default value for columns with cardinality higher than the average value. You might also want to decrease this value when you consistently run profiles on large source files where temporary disk space is low. Default is 5.
Maximum Column Heap Size	The cache size for each column profile mapping for flat files. The default value is optimal for most profiling use cases. You can increase this value to 512 if you want to prevent the Data Integration Service from writing some parts of the intermediate profile results to a temporary disk. To avoid creating multiple mappings, and to prevent an impact on compilation performance, you can combine multiple columns in a single mapping. Default is 64.
Maximum Hadoop Batch Execution Pool Size	The maximum number of Hadoop batch requests that the Data Integration Service can run concurrently. Default is 100.

Profiling Jobs for Enterprise Data Lake

Profiling jobs on Hive tables require Hadoop cluster resources through YARN. The profiling jobs compete for resources with other Enterprise Data Lake Service and Data Integration Service jobs like publish and file upload.

Resource contention can be critical for Enterprise Data Lake. Because incremental profiling is also not supported for Hive tables, Enterprise Data Lake performs profiling for all of the Hive tables every time the Hive scan runs.

To avoid resource contention issues, Informatica recommends running profiling jobs during non-business hours or low demand times.

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